



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year 2024-25 Summer term Exam,
Class : M.Tech. WM

Date : 19/06/2025
Time : 14.30 am to
17:30pm

Subject: Hydraulic Structures
Code : PCC-WM511

Max. Marks : 100
Duration: 3hrs

Assume suitable data if necessary.

Q.1	a	Differentiate between the tail water rating curve(TWRC) and the jump height curve(JHC). How would you select the most suitable type of energy dissipating device for different relative positions of two curves?	08	CO1
	b	Give detailed classification of crest gate with neat sketches	12	CO2
Q.2	a	List and explain various factors affecting coefficient of discharge "C" of an Ogee spillway	08	CO2
	b	Design a non-overflow section of a concrete gravity dam using multi-step method upto IIzone for the following data: Top width = 6.25 m $C = 0.1$, $\xi = 0.5$ Fetch = 40 Kms Wind velocity = 100 km/hr Wind force = 1 KN/m ² Neglect earthquake forces.	12	CO3
Q.3	a	Explain various flood control measures under channel phase	07	CO2
	b	Explain various points to be considered while selecting site for a dam construction.	07	CO1
	c	List advantages and disadvantages of Syphon spillway	06	CO2
Q.4	a	Explain labyrinth weir and parameters affecting its performance.	08	CO3
	b	Design downstream profile of an ogee spillway with the following data. i) Height of spillway crest above river bed =100m ii) Design discharge = 12000m ³ /s iii) Number of spans = 6 iv) Clear distance between piers =15m v) Thickness of pier =3m vi) Slope of downstream face of the overflow section = 0.8H:1V vii) Assume any data if required. Use normal graph present on the last page of answer sheet to plot the Ogee profile.	12	CO4
Q.5	a	A stepped spillway consisting of 9 steps, 1m high and 2.5 m long each, is to pass a unit discharge of 0.3 cumec/m. Assess the type of flow	10	CO4

	conditions on the chute and energy dissipation at the base of the spillway. Hint: Chanson equation for $0.2 \leq h/l \leq 0.6$, is $(y_c/h) \leq 0.0916(h/l)^{-1.276}$ Residual head is given by $\frac{H_r}{H_{\max}} = \frac{0.54(y_c/h)^{0.275} + 1.715(y_c/h)^{-0.55}}{1.5 + (H_d/y_c)}$		
b	List advantages and disadvantages of Syphon spillway.	10	CO3

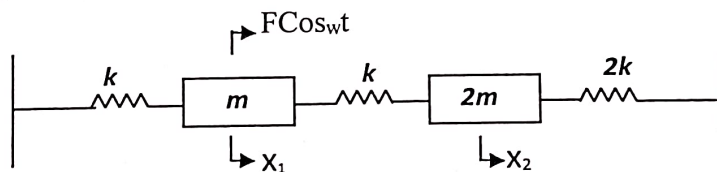


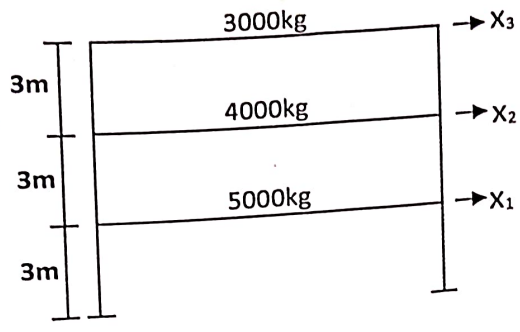
Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Civil Engineering
Summer Term Examination
Academic Year: 2024-25 (Semester – I)

Class: FY M.Tech. (Structural Engineering)	Subject: Structural Dynamic	Subject Code: PCC-SE503
Date: 21.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Non programmable calculator is permitted.

Q.N.			Marks	CO	BT
Q.1	a)	Define :-1) Resonance 2) Degree of freedom 3) Stiffness 4) Mode Shape 5) Damping	05	01	BT1
	b)	Define shear building and state its assumptions.	08	02	BT3
	c)	Explain Energy Method and Rayleigh's method and also derive the equation for the same.	12	03	BT3
Q.2	a)	What is the difference between response of structure to static loading and dynamic loading?	06	02	BT2
	b)	A 1000 kg machine is mounted on four identical springs of total spring constant K and having negligible damping. The machine is subjected to a harmonic external force of amplitude $F_o = 490\text{N}$ and frequency of 180 rpm. Determine the amplitude of motion of machine of maximum force transmitted to the foundation because of unbalanced force when $K=1.96 \times 10^6 \text{ N/m}$.	08	04	BT4
Q.3	a)	Determine the modes of vibration and the steady state response of the given system as shown in fig.1: -  Fig.1	10	05	BT5
	b)	Explain Stodala Method for iteration for determination of fundamental frequency and corresponding mode shape.	12	03	BT3
Q.4	a)	A SDOF system is subjected to sudden Applied load F_o . Find the response by using Duhamel's Integral.	10	04	BT4

	b)	Determine the natural frequencies and mode shapes of the given MDOF system ($EI = 4.5 \times 10^6 \text{ Nm}^2$) as shown in fig.2: -	15	05	BT5
		 Fig.2			
Q5	a)	What is peak ground acceleration? Explain the estimated steps for PGA.	06	03	BT3,6
	b)	What is the shape of design spectrum? Classify the types of time period ranges.	08	02	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	To recall fundamentals of Mathematics and Engineering Mechanics.
CO 2	To Understand the effects of system/model parameters on dynamic response.
CO 3	To apply and formulate dynamic equilibrium equations for SDOF and MDOF systems.
CO 4	To analyze SDOF and MDOF systems using classical and numerical methods.
CO 5	To evaluate and obtain the response of SDOF, MDOF systems and conduct modal analysis of MDOF systems.